Derivation Notes for Cryogenic Device Recovery Model

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Assume the trapped-charge population decays with a first-order rate law:

$$\frac{dQ_{\text{trap}}}{dt} = -\frac{1}{\tau_{\text{rec}}(T)} Q_{\text{trap}}.$$

Integrating gives

$$Q_{\text{trap}}(t,T) = Q_0 \exp\left(-\frac{t}{\tau_{\text{rec}}(T)}\right),$$

where Q_0 is the trapped-charge population immediately after irradiation or immediately after a trapping event.

Now assume the recovery time constant is thermally activated:

$$\tau_{\text{rec}}(T) = \tau_0 \exp\left(\frac{E_a}{k_B T}\right).$$

Substituting into the recovery law gives

$$Q_{\text{trap}}(t,T) = Q_0 \exp\left[-\frac{t}{\tau_0} \exp\left(-\frac{E_a}{k_B T}\right)\right].$$

This makes the low-temperature limit clear. As T decreases, the factor

$$\exp\left(-\frac{E_a}{k_B T}\right)$$

becomes very small, so the effective relaxation slows strongly.

A useful engineering metric is the time to 50% recovery, obtained from

$$\frac{Q_{\text{trap}}}{Q_0} = \frac{1}{2}.$$

This gives

$$t_{50}(T) = \tau_{\text{rec}}(T) \ln 2.$$

Therefore,

$$t_{50}(T) = \tau_0 \ln 2 \;\exp\left(\frac{E_a}{k_B T}\right),$$

which grows rapidly as temperature decreases.